Whole Genome Sequencing: Rapid Proband

Proband Report

BAYLOR GENETICS

DEMOGRAPHIC INFORMATION

PATIENT

NAME:	[redacted]
DATE OF BIRTH:	[redacted]
SEX:	[redacted]
MEDICAL RECORD #:	[redacted]
ACCESSION #:	F00039071006
LAB NUMBER:	[redacted]
FAMILY NUMBER:	[redacted]

TEST INFORMATION

TEST NAME:	Rapid Proband WGS
TEST CODE:	1829
SAMPLE TYPE:	[redacted]
DATE COLLECTED:	[redacted]
DATE RECEIVED:	[redacted]
DATE REPORTED:	09/02/2025

RECIPIENT

PHYSICIAN NAME:	David Horst
FACILITY:	WHHT
LOCATION:	
PHONE:	713-791-7534
FAX:	713-791-7634

ADDITIONAL RECIPIENT

NAME:	Catherine Burson
PHONE:	- -
FAX:	720-754-4906

ADDITIONAL RECIPIENT

NAME:	Margaret Campbell
PHONE:	713-791-7534
FAX:	713-791-7634

CLINICAL INDICATION

Based on the submitted clinical information, the patient has premature birth, feeding difficulties, neonatal hypoglycemia, hypoketotic hypoglycemia, hyperinsulinemia, neonatal respiratory distress, pulmonary opacity, apnea of prematurity, bradycardia, heart murmur, patent ductus arteriosus, patent foramen ovale, anemia, decreased circulating free fatty acid level, hypotension, retinopathy, and abnormality of the urinary system physiology.

RESULTS

NEGATIVE FINDINGS

RESULTS SUMMARY

The analysis did not identify any variant that can explain the patient's clinical phenotype. No causative variants in the PHOX2B gene of interest were detected.

ADDITIONAL FINDINGS

ACMG recommended secondary findings (v3.2; PMID 37347242)

Client has not selected to receive these findings.

Other Incidental Findings

None Detected.

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RECOMMENDATIONS

• Genetic counseling is recommended.
• Contact laboratory for requests regarding reanalysis and raw data.

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QUALITY METRICS

Percentage depth of coverage at 20x: **99.72%**
Mean depth of coverage: **60.72x**

METHODOLOGY

Genomic DNA was fragmented and indexed to prepare individual libraries. The libraries were purified and pooled for sequence analysis on the Illumina NovaSeq platform for paired end reads. Internal quality control measures are performed to ensure sample integrity.

Data analysis and interpretation are performed by the Baylor Genetics analytics pipeline. Data are aligned to the human reference genome build GRCh38 using the Illumina Dragen BioIT Platform. Variant calling is performed using the Illumina Dragen haplotype-based variant calling system and Illumina Dragen genome wide depth based CNV (Copy Number Variants) caller with custom modifications from Baylor Genetics. Short tandem repeat calling is performed using the Illumina Expansion Hunter tool.

The automated genetic interpretation platform from Emedgene is implemented for variant annotation and interpretation with proprietary algorithms and open-source data sets such as gnomAD, EVS (Exome Variant Server), ClinVar, and professional resources such as HGMD (Human Gene Mutation Database) Pro. The variants were interpreted according to ACMG (American College of Medical Genetics) guidelines (reference 1) and patient phenotypes.

Variants are confirmed by Sanger sequencing, specific PCR, Multiplex Ligation-dependent Probe Amplification (MLPA), or another methodology, if sequencing or copy number variants detected by NGS do not meet internal quality standards or are within highly homologous regions.

Synonymous variants, noncoding variants not affecting splicing sites, and common benign variants are excluded from interpretation unless previously reported in the literature as possibly pathogenic variants. This test may not provide detection of certain variants or portions of certain genes due to insufficient sequencing depth, local sequence characteristics, high/low genomic complexity, or the presence of closely related pseudogenes. Small deletions or insertions larger than 25bp, low-level mosaic variants, structural variants such as inversions, and/or balanced translocations may not be detected with this technology.

Variant types being analyzed include sequencing variants and CNV in nuclear genes and mitochondrial DNA (mtDNA), and short tandem repeats (STR). Uniparental disomy (UPD) analysis is done only in a trio setting. Regions of homozygosity (ROH) are reported when greater than 5 Mb. Sensitivity of detection is decreased for mtDNA sequencing variants and CNVs at heteroplasmy levels lower than 5% and 10%, respectively. Pathogenic events of short tandem repeats can be detected within the genomic regions of following genes: AFF2, AR, ATN1, ATXN1, ATXN2, ATXN3, ATXN7, ATXN10, ATNX8OS, C9orf72, CACNA1A, CNBP, CSTB, DIP2B, DMPK, FGF14, FMR1, FXN, GLS, HTT, JPH3, NOP56, NOTCH2NLC, PABPN1, PHOX2B, PPP2R2B, RFC1, TBP, and TCF4. Sensitivity of detection is reduced for borderline or incomplete penetrance alleles.

The proband report contains results of genes related to the patient's clinical phenotype. Variants relevant to the phenotype are reported as positive (pathogenic/likely pathogenic) or variants of uncertain significance (VUS). Benign and likely benign variants are not reported. This interpretation is based on the clinical information provided by the referring institution and our current understanding of the gene and variants at the time of reporting. Additional methodologies may be used to aid in variant interpretation. Results in the report should be carefully correlated to the individual's clinical findings by the referring physician.

If elected, secondary findings including pathogenic and likely pathogenic variants in genes included in the ACMG secondary findings recommendations, or other findings determined to be severe and medically significant (incidental findings), will be reported. The ACMG recommendations (ACMG SF v3.2 (reference 2)) include the following genes: ACTA2, ACTC1, ACVRL1, APC, APOB, ATP7B, BAG3, BMPR1A, BRCA1, BRCA2, BTD, CACNA1S, CALM1, CALM2, CALM3, CASQ2, COL3A1, DES, DSC2, DSG2, DSP, ENG, FBN1, FLNC, GAA, GLA, HFE, HNF1A, KCNH2, KCNQ1, LDLR, LMNA, MAX, MEN1, MLH1, MSH2, MSH6, MUTYH, MYBPC3, MYH11, MYH7, MYL2, MYL3, NF2, OTC, PALB2, PCSK9, PKP2, PMS2, PRKAG2, PTEN, RB1, RBM20, RET, RPE65, RYR1, RYR2, SCN5A, SDHAF2, SDHB, SDHC, SDHD, SMAD3, SMAD4, STK11, TGFBR1, TGFBR2, TMEM127, TMEM43, TNNC1, TNNI3, TNNT2, TP53, TPM1, TRDN, TSC1, TSC2, TTN, TTR, VHL, and WT1. Variants unrelated to the patient's phenotype but potentially clinically significant in genes with no known disease association will be reported if elected.

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LIMITATIONS

The interpretation of nucleotide changes is based on our current understanding of the specific genes. This interpretation may change over time as more information about these genes becomes available. Possible diagnostic errors include sample mix-ups, genetic variants that interfere with analysis, incorrect assignment of biological parentage, or other sources. Please contact a genetic counselor at Baylor Genetics if there is reason to suspect one of these sources of error.

REFERENCES

1. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL; ACMG Laboratory Quality Assurance Committee. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med. 2015 May;17(5):405-24. doi: 10.1038/gim.2015.30. Epub 2015 Mar 5. PMID: 25741868; PMCID: PMC4544753.
2. Miller DT, Lee K, Abul-Husn NS, Amendola LM, Brothers K, Chung WK, Gollob MH, Gordon AS, Harrison SM, Hershberger RE, Klein TE, Richards CS, Stewart DR, Martin CL; ACMG Secondary Findings Working Group. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). Genet Med. 2023 Aug;25(8):100866. doi: 10.1016/j.gim.2023.100866. PMID: 37347242.

DISCLAIMER

This test was developed, and its performance was determined by the Baylor Miraca Genetics Laboratory DBA Baylor Genetics. It has not been cleared or approved by the U.S. Food and Drug Administration (FDA). Since FDA is not required for clinical use of this test, this laboratory has established and validated the test's accuracy and precision, pursuant to the requirement of CLIA'88. This laboratory is licensed and/or accredited under CLIA and CAP (CAP#2109314 / CLIA# 45D0660090; Lab Director: Christine M. Eng, MD).

ELECTRONICALLY SIGNED BY

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